

Problem

You are given an array of N integers representing the weights of people.

Your task is to divide the people into two groups such that:

The two groups have the same number of people, or their sizes differ by at most one. The absolute difference between the total weights of the two groups is minimized.

Submissions

Print the minimum possible difference between the total weights of the two groups.

Input Format

The first line contains an integer N , the number of people. The second line contains N space-separated integers representing their weights.

Leaderboard

Constraints

 $2 \leq N \leq 30$ $1 \leq arr[i] \leq 1000$

Output Format

Print a single integer representing the minimum possible difference between the total weights of the two groups.

Discussions

Sample Input 0

```
4
3 4 5 6
```

Sample Output 0

```
0
```

Sample Input 1

```
5
1 2 3 4 5
```

Sample Output 1

```
1
```

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Language

Python 3



```
1 # Enter your code here. Read input from STDIN. P
2 n = int(input())
3 arr = list(map(int, input().split()))
4 total = sum(arr)
5 need = n // 2
6 ans = total
7 def solve(i, count, s):
8     global ans
9     if count == need:
10         diff = abs(total - 2 * s)
11         ans = min(ans, diff)
12         return
13     if i == n:
14         return
15     solve(i + 1, count + 1, s + arr[i])
16     solve(i + 1, count, s)
17 solve(0, 0, 0)
18 print(ans)
```

Line: 17 Col: 15

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☐ Test against custom input

Test case 0

Test case 1

Compiler Message

Success